Toward a four-dimensional Yang–Mills construction

Notebook 23: gauge covariance of the small-field step

Research working document

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Abstract

We prove local gauge covariance of the actual homogeneous small-field chart integral, uniformly in prior classical refinement. A block-constant lift transports the physical stationary branch and the full nonlinear retained-coordinate pivot chart. Its real adjoint action preserves the retained cutoff exactly. Haar densities, pivot straightening determinants, the separate delta normalization, the centered action remainder, and the Gaussian determinant all transform consistently. The complete generated coarse correction is therefore locally gauge invariant. Its linear jet vanishes for a semisimple gauge group and its quadratic jet satisfies the exact finite-regulator Ward identity. Connected activity bounds imply exponential Hessian summability. Under an additional translation-invariant kernel-limit hypothesis the quadratic multiplier vanishes at zero momentum. The same integral symmetry applies to an inherited invariant functional when that integral is controlled; nonvanishing is explicitly required before taking its logarithm.

The principal inputs are the local classical gauge identity of Notebook 16, the exact physical chart and measure factors of Notebook 19, the controlled classical-reference integration of Notebook 21, and the stationary-background support transport of Notebook 22. All identities below refer to those concrete maps.

1 The common domain and the exact chart integral

Fix a compact connected semisimple matrix group G, a positive invariant color inner product, and real orthonormal color coordinates. The homogeneous current lattice has spacing one, prior classical refinement is $M = L^k$, and the next blocking factor is fixed odd L. Use the source geometry and ordinary averaged contours of Notebooks 19–22. All constants below may depend on L and G, but not on M or compatible periodic volume. The identities concern the established small-potential domains, including their smaller complex continuations. They do not define the classical effective action outside those domains.

Write $V = e^{ia}$ for a current background and $W = e^{iw}$ for next-coarse data. The physical branch $V_*(W)$ satisfies

$$M_L(V_*(W)) = W, \qquad P_{V_*(W)}(y, x) = 1. \tag{1}$$

Notebook 22 identifies it with the unique small stationary solution of the classical action plus the actual absolute averaged-contour gauge penalty. Its strict support series is also available there. The action S_M is the ordinary classical constrained minimum of the original Wilson action, with the coupling excluded.

For any sufficiently small independent background V, let

$$F_V(B; c) = \frac{1}{i} \log\Big[M_L(e^{iB}V)(c) M_L(V)(c)^{-1}\Big]. \tag{2}$$

The exact nonlinear pivot chart $\Phi_V(z)$ is characterized by

$$F_V(\Phi_V(z)) = 0, \qquad \Phi_V(z)|_{\text{retained}} = z. \tag{3}$$

Each constraint reads its own pivot and no other pivot. This gives uniqueness of its small solution with prescribed retained coordinates. Denote its tangent insertion by $K_V = D\Phi_V(0)$ and its pivot derivative at zero by

$$A_{c,p}(V) = D_{p(c)}F_V(0;c), \qquad \mathcal{D}(V) = \prod_c \det A_{c,p}(V)^{-1}. \tag{4}$$

The real determinants have the positive orientation fixed at the identity background.

Let $\mathscr{G}$ be the absolute exponential gauge action and put

$$U_V(z) = e^{i\Phi_V(z)}V,$$

$$T_V(z) = S_M\left(\frac{1}{i}\log U_V(z)\right) + \mathscr{G}(U_V(z)), \tag{5}$$

$$A_M(V) = D_z^2 T_V(0), \qquad A_{M,0} = A_M(1). \tag{6}$$

Thus A_M includes both the constraint curvature and the logarithmic coordinate curvature in the classical action. Away from the physical branch the gauge Hessian is also its full second derivative. These are the analytic centered product-domain extensions used in Notebooks 20–21, not a Hessian obtained by dropping a background current term.

Subtract the entire Taylor polynomial through degree two:

$$\mathcal{R}_V(z) = T_V(z) - T_V(0) - DT_V(0)z - \tfrac{1}{2}\langle z, A_M(V)z\rangle. \tag{7}$$

Let $j(X)$ be the Haar exponential-coordinate density divided by its value at zero, and define the relative measure factor

$$J_V(z) = \prod_b j(\Phi_V(z)(b)) \prod_c \frac{\det A_{c,p}(V)}{\det D_{p(c)}F_V(\Phi_V(z);c)}. \tag{8}$$

It is exactly the relative Haar density times the ambient straightening determinant, as proved by pivot elimination in Notebook 19. The rectangular derivative $D\Phi_V$ is not used as a square Jacobian.

For positive real g, use the fixed retained window

$$\chi_R(\xi) = \prod_{b \text{ retained}} \mathbf{1}_{\{|\xi(b)|_{\mathfrak{g}} \le R\}}. \tag{9}$$

These are invariant Euclidean color balls at each retained bond, not a product of intervals in individual color coordinates. A scalar complex polydisc used for a Cauchy estimate is a different object. The normalized integral is

$$Z_{M,g}(V) = \int d\mu_{A_M(V)^{-1}}(\xi)\, \chi_R(\xi)\, e^{-g^{-2}\mathcal{R}_V(g\xi)} J_V(g\xi). \tag{10}$$

The real Gaussian is normalized. For complex backgrounds its density is continued on the same real integration variables, with the determinant branch fixed at the real positive reference. The bounds of Notebook 21 make this analytic and nonvanishing on a common small product domain, for its stated small-coupling regime. At the physical branch $DT_V(0) = 0$ and $\mathscr{G}(V) = 0$; this integral therefore supplies precisely the correction in the exact small-field chart formula.

2 Gauge covariance of the stationary background and pivot chart

For a next-coarse site gauge $u(y)$, its block-constant current lift is

$$u^{\uparrow}(x) = u(y), \qquad x \in B_L(y). \tag{11}$$

Use the convention $V^v(b) = v(b_-)V(b)v(b_+)^{-1}$ and the analogous convention for W. We restrict W and u so all transformed fields, comparison branches, and the gauge paths used below stay in one common small domain. For example, the current bound $\|\log V^{u^{\uparrow}}/i\|_\infty \le C(\|a\|_\infty + \|\log u/i\|_\infty)$ and the prior uniform branch bounds give a fixed smaller choice of radii.

Lemma 2.1 (Block lift and stationary branch)**.** *For such real fields and gauges,*

$$\begin{aligned} P_{V^{u\uparrow}}(y,x) &= u(y)P_V(y,x)u(y)^{-1}, \\ M_L(V^{u^{\uparrow}}) &= M_L(V)^u, \qquad V_*(W^u) = V_*(W)^{u^{\uparrow}}. \end{aligned} \tag{12}$$

These identities continue holomorphically in the corresponding smaller complex parameter domain.

Proof. Every internal root contour lies in one block. Its endpoint covariance is therefore conjugation by the same $u(y)$, and the analytic group mean defining P has that covariance. The ordinary coarse average has endpoint covariance by its construction.

The local gauge identity for S_M was proved in Notebook 16 by two minimum comparisons, using a smooth original-fine extension of arbitrary current-site gauge data. It applies to $u^{\uparrow}$: the input is specified at current spacing one, and the smooth extension has a norm bounded by the supremum of those site data uniformly in M. A jump between current blocks does not produce an M-dependent derivative in that extension. Also $\mathscr{G}(V^{u^{\uparrow}}) = \mathscr{G}(V)$, since the contour matrices are conjugated and its defining trace is invariant.

Consequently the gauge diffeomorphism transports constrained stationarity of $S_M + \mathscr{G}$ at $V_*(W)$ to constrained stationarity at $V_*(W)^{u^{\uparrow}}$ with data W^u. Its internal contours equal one by the first identity, and the common-domain restriction places it in the same uniqueness neighborhood as $V_*(W^u)$. The strictly positive retained Hessian and stationary uniqueness established in Notebook 22 identify the two configurations. No assertion that every stationary configuration has $P = 1$ is needed. All the maps are holomorphic on the smaller complex domains. Equality on an open real form extends coordinate by coordinate to the connected complex parameter neighborhood. □

Put $V' = V^{u^{\uparrow}}$ and define the full relative bond action and its retained restriction by

$$(R_u B)(b) = \mathrm{Ad}_{u^{\uparrow}(b_-)}\, B(b), \qquad D_u = R_u|_{\text{retained}}. \tag{13}$$

For real compact gauges these maps are orthogonal in each color fiber and have determinant one on their respective coordinate spaces. Indeed an orthogonal adjoint matrix has determinant in $\{1,-1\}$, and connectedness of G fixes it to one.

Lemma 2.2 (Exact nonlinear chart equivariance)**.** *On the common chart,*

$$F_{V'}(R_u B; c) = \mathrm{Ad}_{u(c_-)}\, F_V(B; c), \tag{14}$$

$$\Phi_{V'}(D_u z) = R_u \Phi_V(z), \qquad K_{V'} D_u = R_u K_V. \tag{15}$$

Proof. One has $e^{iR_uB}V' = (e^{iB}V)^{u^{\uparrow}}$. Both ordinary averages in the relative product defining F transform with the same endpoint gauges. Their product is conjugated at the initial coarse endpoint. Conjugacy of the small logarithm proves (14).

The candidate $R_u\Phi_V(z)$ solves the transformed constraint and has retained coordinates $D_u z$. The unique small pivot solution is therefore $\Phi_{V'}(D_u z)$. Differentiating this identity at zero gives the tangent identity. □

3 Covariance of every normalization factor

Proposition 3.1 (Exact invariance of the normalized integral)**.** *For real small u and common backgrounds as above,*

$$A_M(V') = D_u A_M(V) D_u^T, \qquad \mathcal{D}(V') = \mathcal{D}(V), \tag{16}$$

$$Z_{M,g}(V') = Z_{M,g}(V). \tag{17}$$

Their analytic identities extend to the stated common complex domains.

Proof. Action gauge invariance, conjugacy of the absolute gauge action, and Lemma 2.2 imply

$$T_{V'}(D_u z) = T_V(z). \tag{18}$$

The constant, linear, and quadratic Taylor parts transform separately under the fixed linear map D_u. Thus

$$\mathcal{R}_{V'}(D_u z) = \mathcal{R}_V(z). \tag{19}$$

The quadratic identity reads $D_u^T A_M(V') D_u = A_M(V)$, which gives the first formula in (16). This argument also applies to the centered extension at independent backgrounds that are not stationary.

Differentiating (14) in the pivot of c gives, at a general small B,

$$D_{p(c)} F_{V'}(R_u B; c) = \mathrm{Ad}_{u(c_-)}\, D_{p(c)} F_V(B; c)\, \mathrm{Ad}^{-1}_{u^\uparrow(p(c)_-)}\,. \tag{20}$$

Both outside factors have determinant one. Evaluating at $B = 0$ proves invariance of $\mathcal{D}(V)$. Evaluating at $\Phi_V(z)$ proves invariance of each normalized straightening determinant in (8). Haar measure is invariant under conjugation, and $\det \mathrm{Ad}_u = 1$ in these real coordinates, so $j(\mathrm{Ad}_u X) = j(X)$. Together with the chart identity this proves

$$J_{V'}(D_u z) = J_V(z). \tag{21}$$

Make the real change of retained variables $\xi' = D_u \xi$. It has unit Jacobian, carries the normalized Gaussian with precision $A_M(V)$ to the one with precision $A_M(V')$, and preserves every color norm in (9). Equations (19) and (21) prove (17). This uses a real orthogonal change of variables; it neither complexifies the sharp indicator nor deforms its integration domain.

Each scalar identity is holomorphic in the background and gauge parameters on a smaller common domain. The real-form identity theorem then supplies the complex identity. For complex adjoint matrices the relevant algebraic transpose is bilinear; no claim of unitary invariance of a complex sharp window is made. □

Define the separate Gaussian correction and total quantum correction by

$$\mathcal{Q}_M(V) = -\tfrac{1}{2} \log \det\Big(A_M(V) A_{M,0}^{-1} \Big)\,, \tag{22}$$

$$\mathcal{E}^{\mathrm{eff}}_{M,g}(W) = \log \frac{\mathcal{D}(V_*(W))}{\mathcal{D}(1)} + \mathcal{Q}_M(V_*(W)) + \log Z_{M,g}(V_*(W)). \tag{23}$$

All logarithms use compatible analytic branches. The determinant branches are fixed by positivity and the identity reference; the integral branch is the convergent connected expansion from Notebook 21. On real sufficiently small fields that integral is strictly positive, so this branch agrees with its real logarithm.

Theorem 3.2 (Local gauge invariance of the full correction)**.** *In the common small domain,*

$$\mathcal{E}^{\mathrm{eff}}_{M,g}(W^u) = \mathcal{E}^{\mathrm{eff}}_{M,g}(W). \tag{24}$$

The correction includes the pivot delta normalization and Gaussian determinant, as well as the fluctuation integral. The explicit field-independent prefactor and classical stationary action in the full chart formula remain separate.

Proof. Lemma 2.1 identifies the two backgrounds. Proposition 3.1 proves invariance of $\mathcal{D}$, $\det A_M$, and $Z_{M,g}$ there. The fixed scalar denominator $\det A_{M,0}$ is unchanged; the matrix $A_{M,0}$ itself need not be invariant under a local gauge. The compatible real logarithms consequently agree. Holomorphic uniqueness gives the stated complex identity. □

Notebook 22 expresses the sum in (23) by strictly supported coarse activities after reducing to a common positive support weight and the stated smaller analytic radius. The theorem proves invariance of that total sum. An arbitrary chosen individual activity, or the deterministic connecting tree used to assign its carrier, need not be gauge invariant.

Proposition 3.3 (An inherited invariant functional)**.** *Suppose an inherited current functional $\mathcal{E}(U)$ is holomorphic and gauge invariant on the needed common domain, and insertion of $e^{\mathcal{E}(U_V(g\xi))}$ in (10) gives a convergent integral $Z^{\mathcal{E}}_{M,g}(V)$. Then*

$$Z^{\mathcal{E}}_{M,g}(V^{u^\uparrow}) = Z^{\mathcal{E}}_{M,g}(V). \tag{25}$$

If additionally this integral is nonvanishing and a compatible holomorphic logarithm is chosen on the connected common domain, the corresponding total logarithmic correction is locally gauge invariant.

Proof. The chart identity gives $U_{V'}(D_u z) = U_V(z)^{u^\uparrow}$, so the inherited factor is unchanged in the same real change of variables. This proves the integral identity. For the logarithm, the possible difference is locally a constant in $2\pi i\mathbb{Z}$ and is zero at the identity gauge; continuous paths in the stated common branch therefore give equality. A verified convergent cluster expansion is one way to supply the separate nonvanishing hypothesis. □

The additional hypothesis is material: convergence of a complex integral alone does not exclude zeros. This proposition supplies symmetry once the inherited integral is controlled. It does not supply its required activity norm or repeated-step bounds.

4 A finite-regulator Ward identity and its kernel consequence

Let

$$F(w) = \mathcal{E}^{\mathrm{eff}}_{M,g}(e^{iw}), \tag{26}$$

or another holomorphic total coarse functional with the same local gauge invariance. Here w is the next-coarse link angle in next-coarse unit coordinates.

Theorem 4.1 (Quadratic Ward identity)**.** *On every admissible finite periodic regulator,*

$$DF(0) = 0. \tag{27}$$

For the signed unit coarse gradient $(\partial\lambda)(b) = \lambda(b_-) - \lambda(b_+)$, one has

$$D^2F(0)[v, \partial\lambda] = 0 \quad \textit{for every } v, \lambda, \qquad H_F\partial = 0, \quad H_F = D^2F(0). \tag{28}$$

Proof. Constant site gauges act by adjoint conjugation on each bond angle. Differentiating their invariance at $w = 0$ says that each individual linear color covector is adjoint invariant. Infinitesimally such a covector annihilates every commutator. Since $\mathfrak{g} = [\mathfrak{g}, \mathfrak{g}]$ for a semisimple Lie algebra, it is zero. Independent variations at each bond prove (27). Only constant gauges in a neighborhood of the identity are needed for this reasoning.

For a site field λ set $u_t = e^{it\lambda}$ and denote the infinitesimal gauge velocity of the potential by $X_\lambda(w) = \frac{d}{dt} w^{u_t}\Big|_{t=0}$. Local multiplication gives $X_\lambda(0) = \partial\lambda$. Gauge invariance states $DF(w)X_\lambda(w) = 0$. Differentiating in the direction v at zero yields

$$D^2F(0)[v, \partial\lambda] + DF(0)(DX_\lambda(0)v) = 0.$$

The second term vanishes by (27), proving (28). This argument concerns the total functional, not the separate members of a support decomposition. □

For completeness, the transported activity norm gives an actual exponentially summable Hessian. It is not necessary to assume this kernel bound independently at a finite regulator.

Lemma 4.2 (Hessian from connected analytic activities)**.** *Suppose $F(w) = \mathrm{Vac} + \sum_K F_K(w|_K)$ with nonempty connected carriers, d_0 scalar coordinates per unit cell, holomorphy on a scalar polydisc of radius r, and*

$$\sup_i \sum_{K \ni i} e^{a|K|} \|F_K\|_r \le \epsilon, \qquad a > 0. \tag{29}$$

For any $0 \le \mu < a$, in the maximum-coordinate block operator norm,

$$\sup_i \sum_j e^{\mu d(i,j)} \|(H_F)_{ij}\| \le \frac{2d_0\epsilon}{r^2}\, C_{a-\mu}, \qquad C_\rho = \sup_{k \ge 1} k e^{-\rho k} \le \frac{1}{e\rho}. \tag{30}$$

An equivalent Euclidean block norm changes only the fixed dimension constant.

Proof. Coordinatewise Cauchy bounds a second scalar partial derivative of F_K at zero by $2r^{-2}\|F_K\|_r$; the factor two covers repeated differentiation in the same coordinate. Its (i, j) block has norm at most $2d_0 r^{-2}\|F_K\|_r$ and vanishes unless $i, j \in K$. Connectedness gives $d(i, j) \le |K| - 1$. Summing its at most $|K|$ column cells and using $|K|e^{\mu(|K|-1)} \le C_{a-\mu}e^{a|K|}$ proves (30). The positive bound justifies the derivative and support sums. □

The bounds of Notebook 22 supply (29) for the total correction on its smaller analytic domain. This gives a uniform Hessian bound at fixed g, L, r and the allowed classical depth. The factor r^{-2} is explicit; this is not a coupling-flow estimate.

Proposition 4.3 (Zero-momentum consequence under a kernel limit)**.** *Assume additionally that a sequence of these Hessians is translation invariant and converges, on tori exhausting all coordinate directions, to a convolution kernel on $\mathbb{Z}^4$, with a uniform positive exponential summability bound. Then its Fourier multiplier $\widehat{H}_F(p)$ is analytic near real momentum zero and*

$$\widehat{H}_F(p)d(p) = 0, \qquad d_\nu(p) = 1 - e^{ip_\nu}, \qquad \widehat{H}_F(0) = 0. \tag{31}$$

The first equation includes the identity on color space.

Proof. Exponential summability gives locally uniform convergence of the Fourier series in a complex strip. The finite-volume Ward identity passes to the limiting kernel; since the gradient is a finite difference operator, this also follows directly by taking limits in its kernel equation. Fourier transformation gives the first identity for continuous real p. For $p = te_\nu$, with $t \neq 0$ small, the only nonzero component of $d(p)$ is $1 - e^{it}$ in direction ν. Thus $\widehat{H}_F(te_\nu)e_\nu = 0$ on every color vector. Continuity as $t \to 0$ annihilates every column direction of $\widehat{H}_F(0)$, proving the last identity. □

The conclusion excludes a nonzero momentum-independent quadratic mass coefficient in that limit. Finite discrete momenta alone would not justify the continuous-momentum step. The additional symmetry and kernel-limit hypotheses have therefore been retained explicitly.

5 Use in the existence argument

The complete defined small-field step preserves local gauge invariance, including its cutoff and normalization factors. Together with the strict coarse support construction, this gives an analytic local correction whose total quadratic kernel obeys the Ward identity. The inherited-functional proposition shows how the same symmetry can pass through a later integration when its analytic norm and nonvanishing are established.

The next task is to extract the vacuum and marginal gauge-action coefficient and control the remaining interactions under repeated scaling. Transversality does not determine that coefficient or its flow. Neither does it alone classify all allowed momentum terms: the rotation and reflection properties used in such a classification must be proved for the actual construction. Stable analytic or gauge-orbit domains, inherited fluctuation-interaction estimates, large-field complements, and observable insertion bounds remain part of the full existence proof. This notebook proves the symmetry input to that analysis.

References

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